THIEN TRANDINH, MASc

Software Engineer

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SKILLS

Languages & Frameworks: **Typescript, C# , .NET, SQL, , Javascript, Unity**
Tooling & Delivery: **Git, CI/CD, Jira, Confluence, Miro, Growthbook (A/B Testing), Agile, Scrum**
Systems & Product: **RESTful API's, Scalable Solutions, Cross-Functional Collaboration, Entity Component System (ECS)**

JOB EXPERIENCE

Prodigy Education, *Software Engineer* • *C#, Typescript • Unity* *Feb 2025 – Jan 2026*

- Main contributor to **Prodigy Math'**s Replay Campaign, a post-game progression system
  - Built modular components (prioritizing SOLID principles) for **ECS (Entity Component system)** that **reduced content iteration time by 50%** and enabled **designer-owned updates**, resulting in new content **increasing subscription conversion by 8.6% and monthly revenue by 11%**
  - Performed cost-analysis of time spent on bugs related to legacy code vs reworking it with new re-usable components to reduce maintenance costs, then implemented these improvements
- **Designed and delivered the Linear Campaign onboarding system**
  - **Reduced new-player churn by 15% and improved subscription conversion by 4%**
  - Validated via A/B testing and **staged rollouts for different cohorts** and eventual legacy-player inclusion
- Assigned as the **Core engineer on Prodigy's rapid-prototyping Game Island initiative**, shipping interactive, monetized experiences at production scale across web and Chromebook platforms
  - **Built A/B-tested monetization**, including CDN video ads optimized for low-end Chromebooks
  - Led development of Neek's Bubble Rescue and Academy Defense, integrating education, progression, and monetization frameworks contributing to increasing player retention by 8%
- Built internal designer & QA tooling
  - **Spreadsheet-to-JSON pipelines reduced data iteration time by 90%**
  - Debug commands and cheat tools significantly improved QA throughput and regression testing
- Built and standardized **AI-assisted engineering workflows** (Codex) to improve onboarding speed, refactoring safety, and test coverage across large codebases
- Partnered closely with designers and product stakeholders to iteratively refine core user flows through rapid prototyping and feedback loops
- Owned multiple live, user-facing features end-to-end, including onboarding flows, interactive content modules, and tutorial systems, and acted as **release owner responsible for production stability and hotfixes**

Gatarn Games Ltd, *Fullstack Software Engineer* • *C# .NET, Typescript, REST, SQL* *Dec 2022 – Dec 2024*

- **Led development of Skillfite.io**, a web-based real-time application with **1000+ daily active users**
- Built and ran live-ops systems and content pipelines **driving an 80% increase** in monthly revenue
- Built the backend platform powering player progression, monetization, and real-time gameplay
- Designed login, and session services connecting web clients to game servers via **TCP and WebSockets**
- Implemented a **SQL-backed player data platform** to support accounts, inventory, and progression
- Architected and optimized the game's core runtime systems, including:

- **Reducing client-server network traffic by ~90%,** improving latency and lowering server costs
- Built a reusable ECS-style architecture to enable modular, data-driven feature development and rapid iteration
- Implementing an **object-pooling framework improving CPU and memory performance by ~80%**

- Built the TypeScript web client powering gameplay, UI, and monetization
  - Designed **reusable, SDK-agnostic ad-integration** layers supporting multiple monetization providers
  - Built extensible UI frameworks enabling rapid feature and UX development
- Drove performance, reliability, and developer velocity across the live service
  - Used Visual Studio and runtime profiling to identify bottlenecks**, improving performance by ~50%**
  - Integrated AI-assisted development workflows to accelerate refactoring, test generation, and large-scale system changes while maintaining production-grade code quality
- Led engineering delivery and team growth
  - Ran standups, sprint planning, sprint reviews, and retrospectives using Jira
  - **Designed and ran the technical hiring process**, interviewing and onboarding new engineers

McMaster Children's Hospital*, Software Engineer* • C#, Unity *Sept 2019 – Nov 2021 | Feb 2026 – current*

- Designed, developed, and maintained a VR-based rehabilitation application currently used in hospital clinics
- **Interacted with Doctors and external stakeholders** to derive requirements and constraints
- Leveraged **Google-Maps API to dynamically obtain terrain data**
  - Designed tuning methods to transform terrain data into a traversable game map
- Maintain and update application per doctor's and physio's requests

Imagine Communications*, Software Engineer* • *Typescript, C# .NET, VUE* *Nov 2021 – Dec 2022*

- Designed **containerized web pages** using VUE.js, Typescript, and Docker
- Worked in **SAFe Agile environment** using Git, Bitbucket, Jenkins and Jira
- Improved customer usability with dynamic landing page, and custom table filtering
- Improved code maintainability by taking an active role in bi-weekly merges of long-term branches

Viral Staging*, Freelancer Developer* • *JavaScript* *May 2022 – Aug 2022*

- Implemented an embeddable HTML/JavaScript geolocation widget, resolving user postal codes to the nearest of **700+ vendors**
- Built **Google Maps+Drive API integrations** to dynamically display location data on customer-facing websites

Evertz*, Software Design Engineer Internship* • JavaScript *May 2017 – Aug 2018*

- Developed and maintained UI control applications, and implemented **third party hardware integration**

EDUCATION & CERTIFICATIONS

McMaster University, Hamilton, Ontario

Software Engineering MASc. (Master of Applied Science) (Link to thesis) *2019 –2022*
Bachelor of Software Engineering *2014 – 2019*

Certifications: *Microsoft Azure Fundamentals (AZ-900)* *2024*